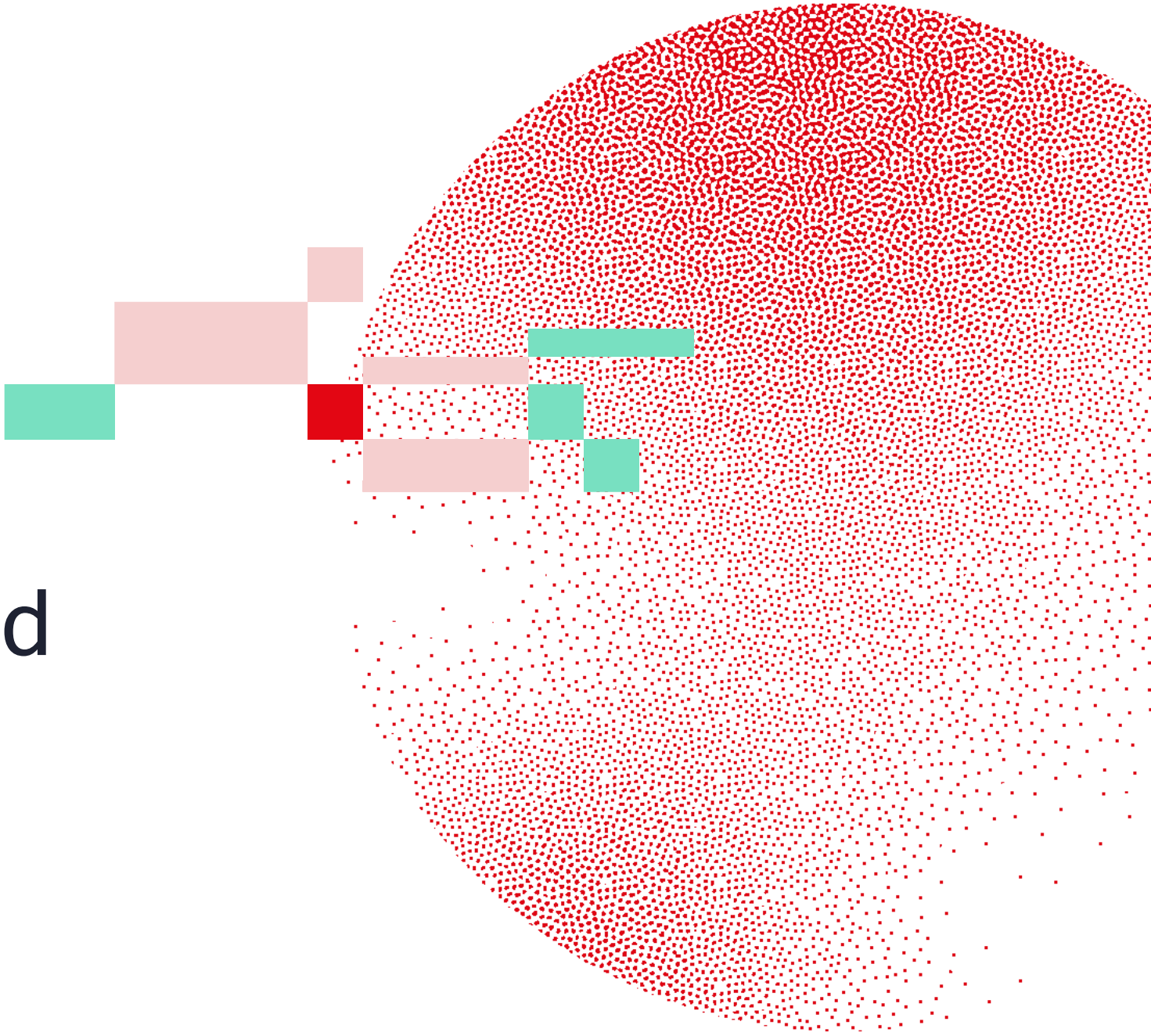


Swiss Institute of Bioinformatics



SINGLE-CELL TRANSCRIPTOMICS WITH R

Normalization, Transformation, and scaling

Deepak Tanwar

November 11-13, 2026

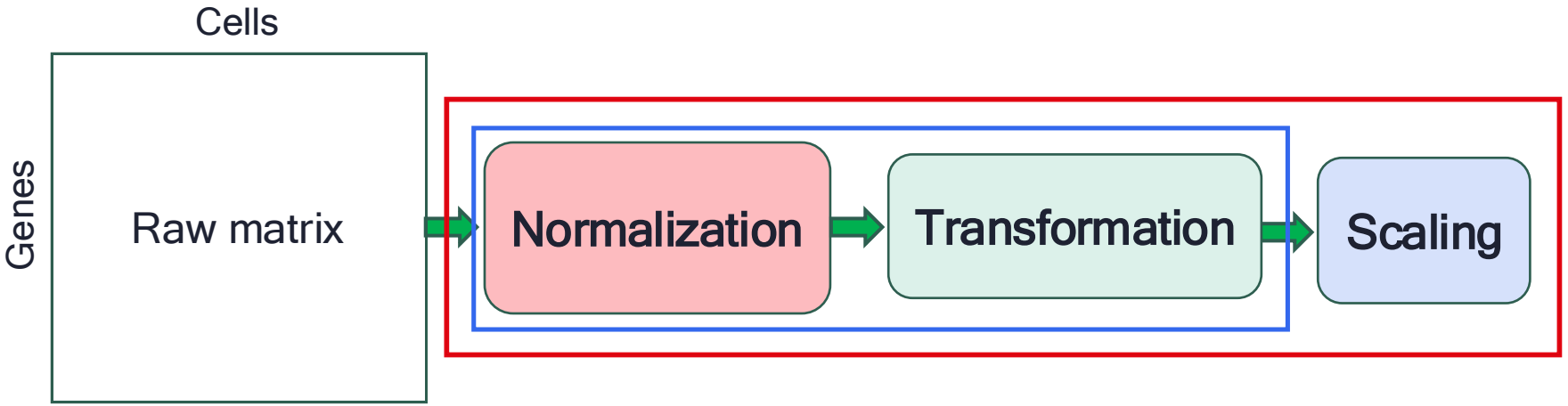
elixir SWITZERLAND

Learning objectives

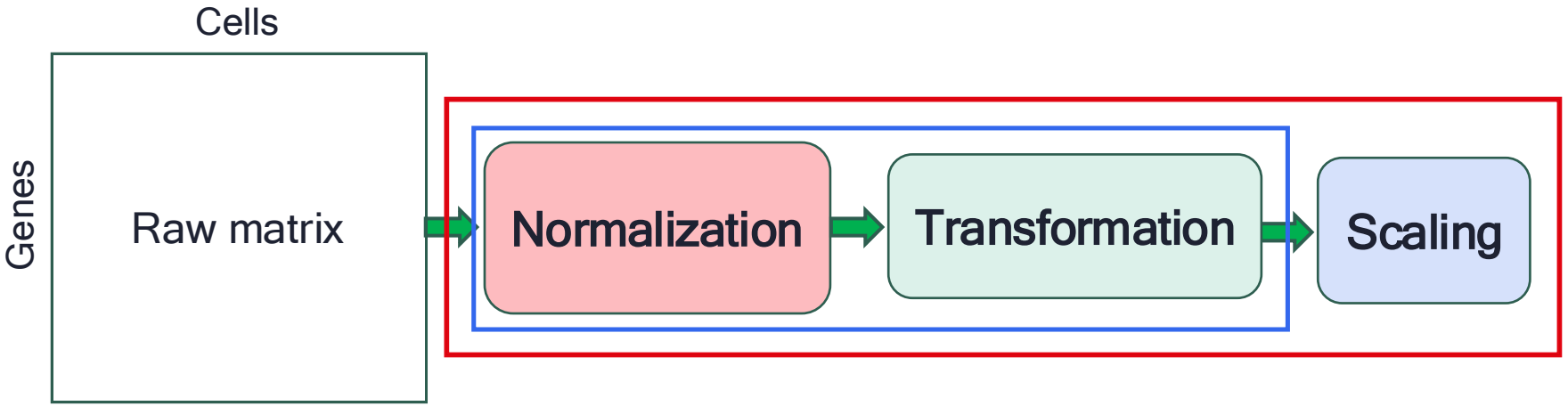
Distinguish Normalization, Transformation, and Scaling

Identify and apply Normalization techniques

Normalization and scaling

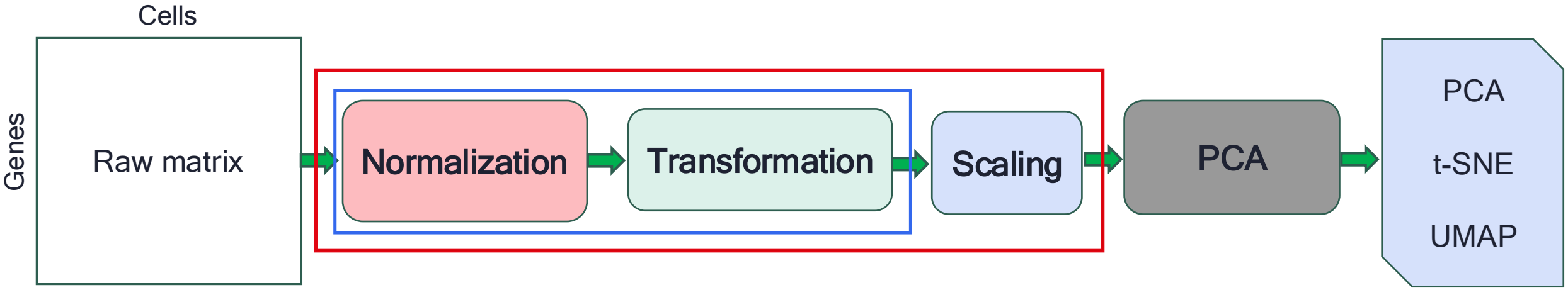


Normalization and scaling

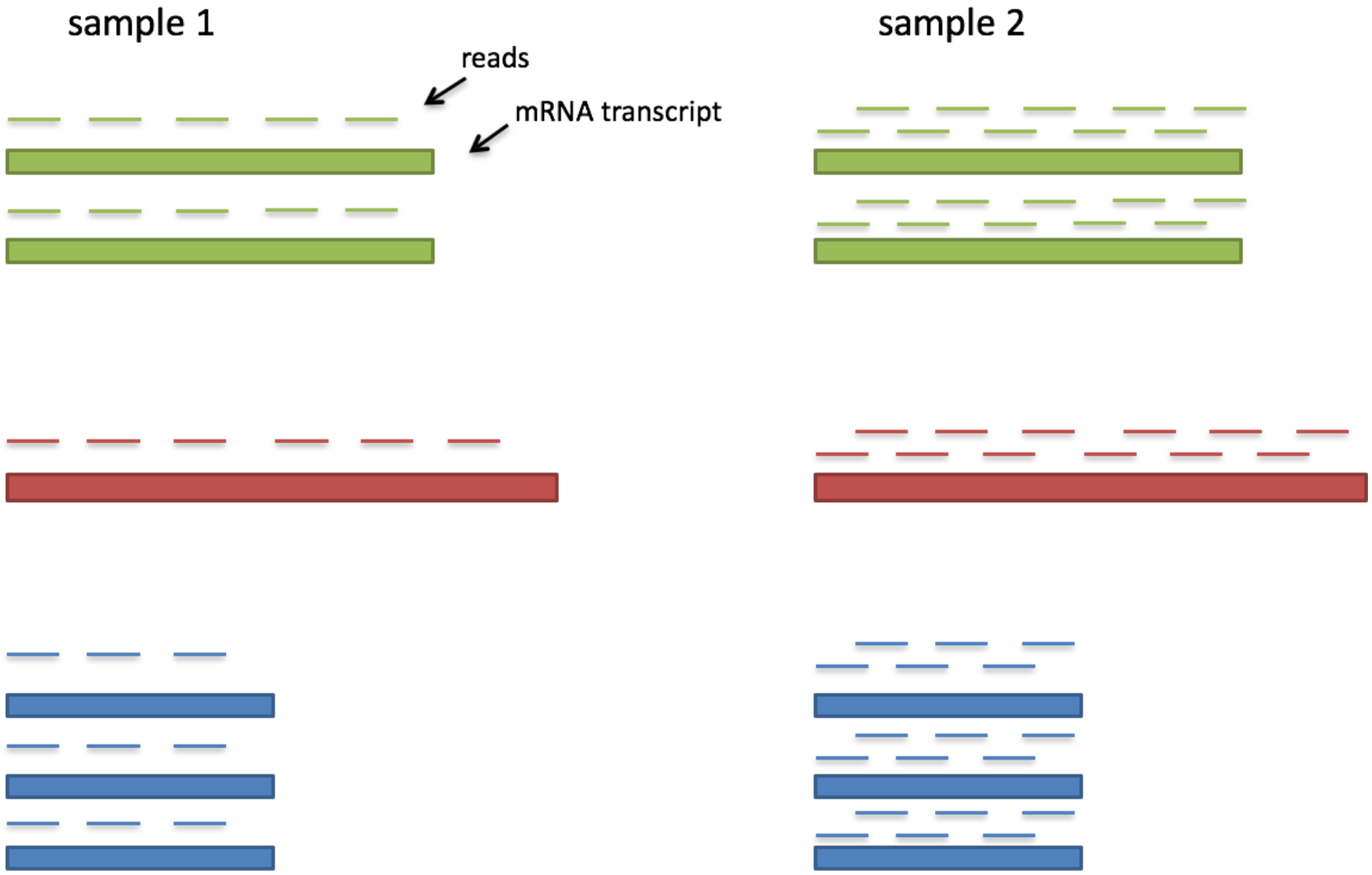


Why am I doing this?

Normalization and scaling



Differences from sequencing data



Problems with raw counts

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1. Library size bias → deeper sequenced cells dominate

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2. Zero-inflation → most genes = 0

Problems with raw counts

1. Library size bias → deeper sequenced cells dominate
2. Zero-inflation → most genes = 0
3. High dynamic range

Goal of normalization and transformation

1. Remove sequencing-depth (library-size) differences so that cells are comparable.

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2. Transform values with `log`

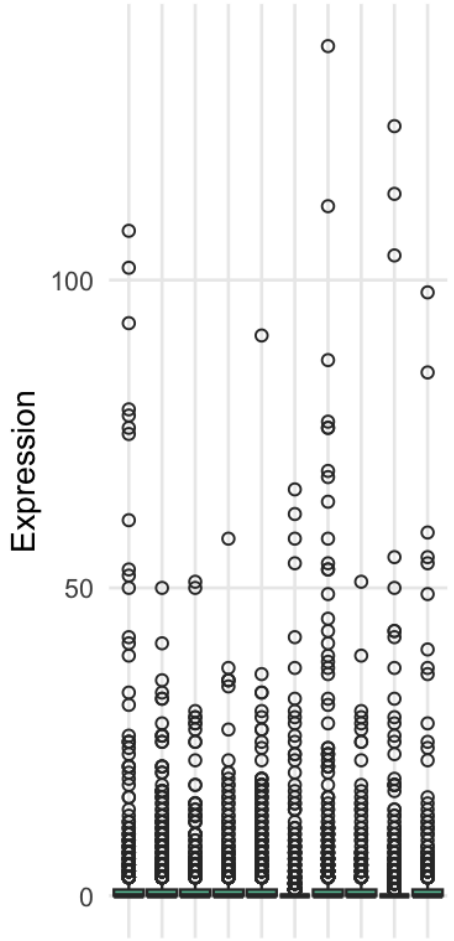
Goal of normalization and transformation

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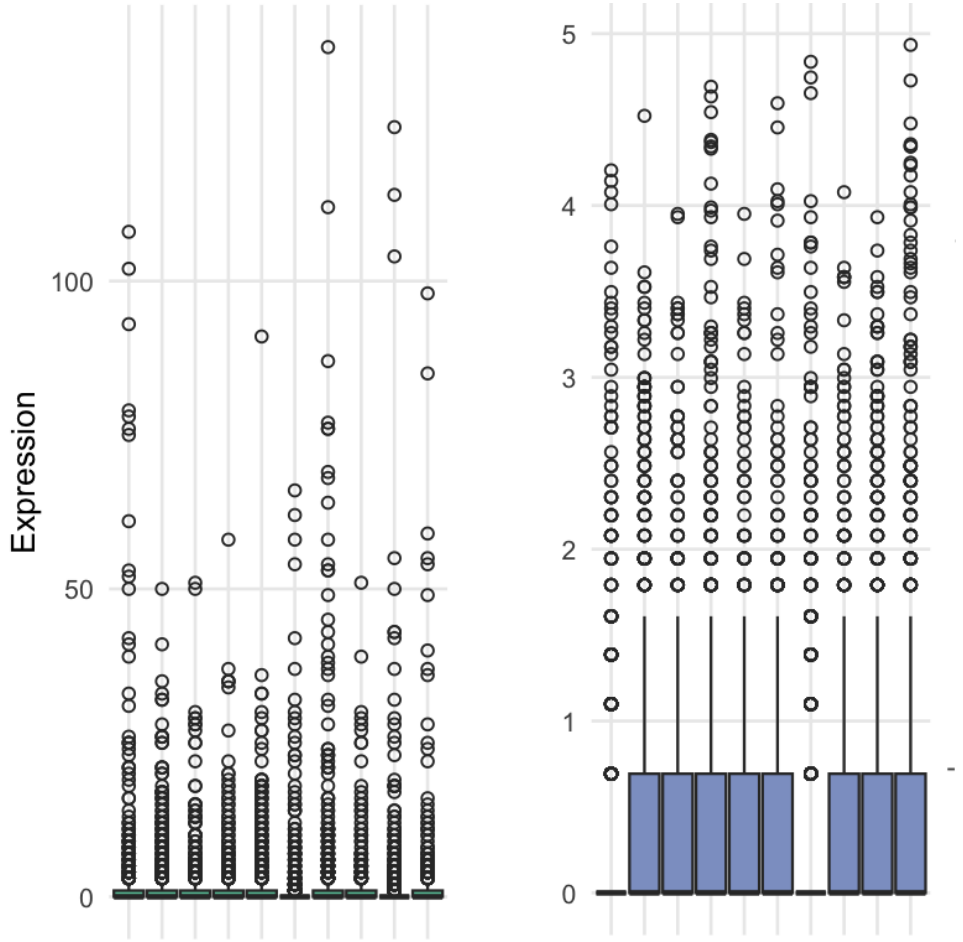
2. Transform values with `log`

3. Multiply each UMI count by 10,000 to make it in a readable/manageable scale

Transformation

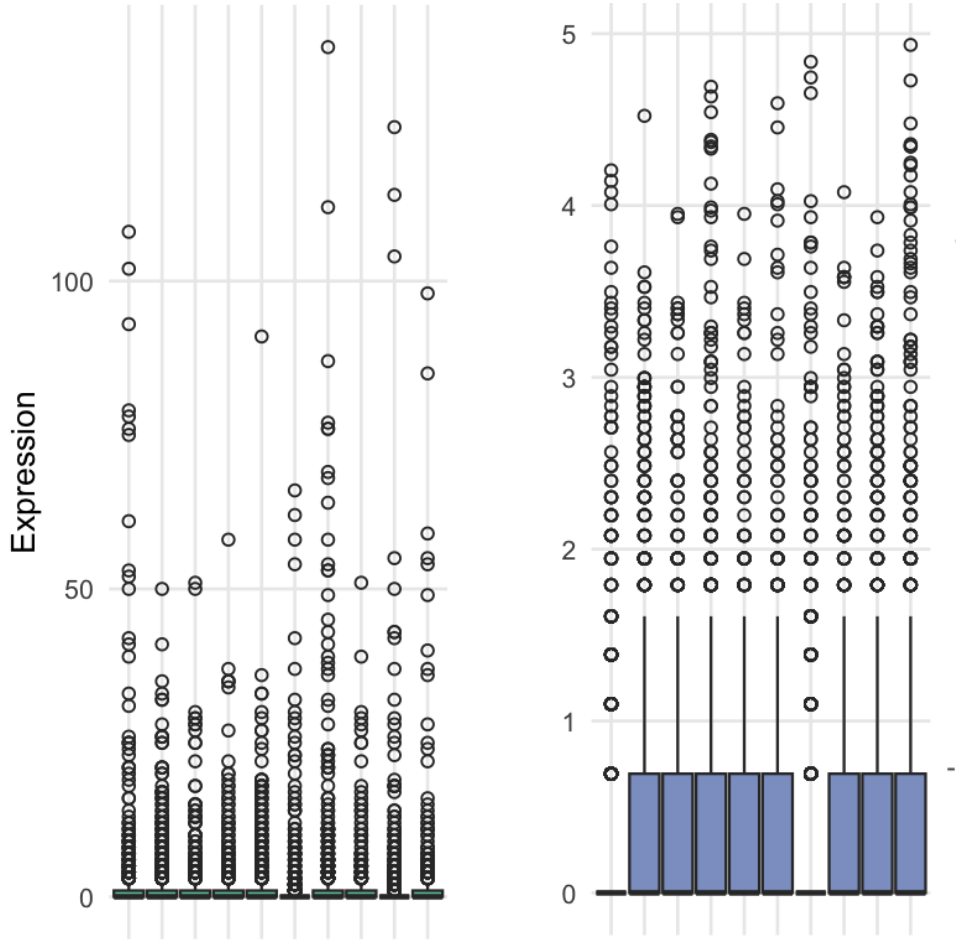


Transformation



$$\text{norm}_{g,c} = \log\left(\frac{\text{count}_{g,c}}{\sum_g \text{count}_{g,c}} \times 10{,}000 + 1\right)$$

Transformation



`Seurat::NormalizeData()`

$$\text{norm}_{g,c} = \log\left(\frac{\text{count}_{g,c}}{\sum_g \text{count}_{g,c}} \times 10{,}000 + 1\right)$$

What Is Scaling?

The Problem: Genes Have Different Scales

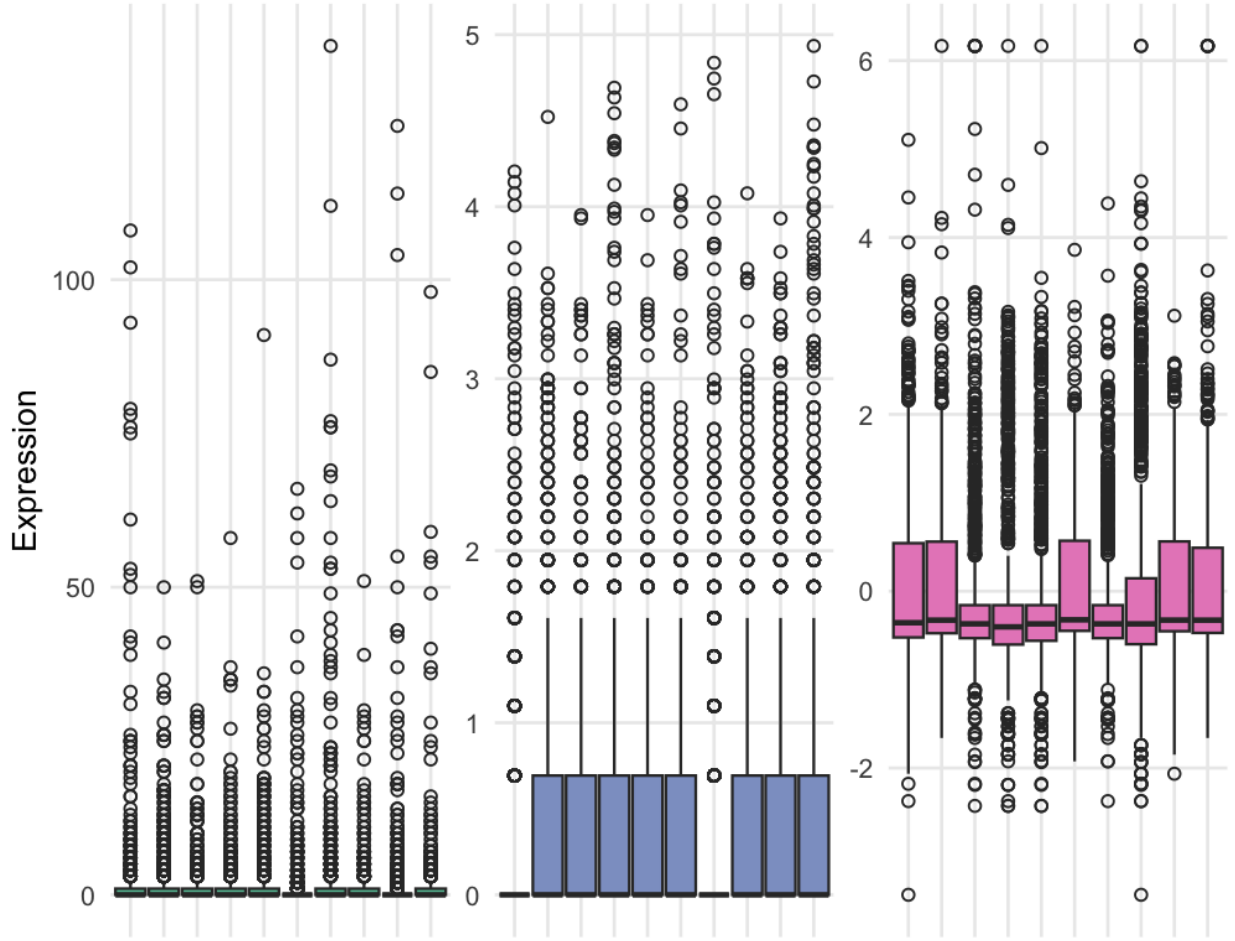
Gene	Mean	SD	Dominates PCA?
LYZ	5	3	YES
CD79A	0.1	0.2	NO

High-expression genes dominate PCA — even if low variance!

Solution: Z-Score Scaling

(Every gene → **mean = 0, SD = 1**)

Scaling



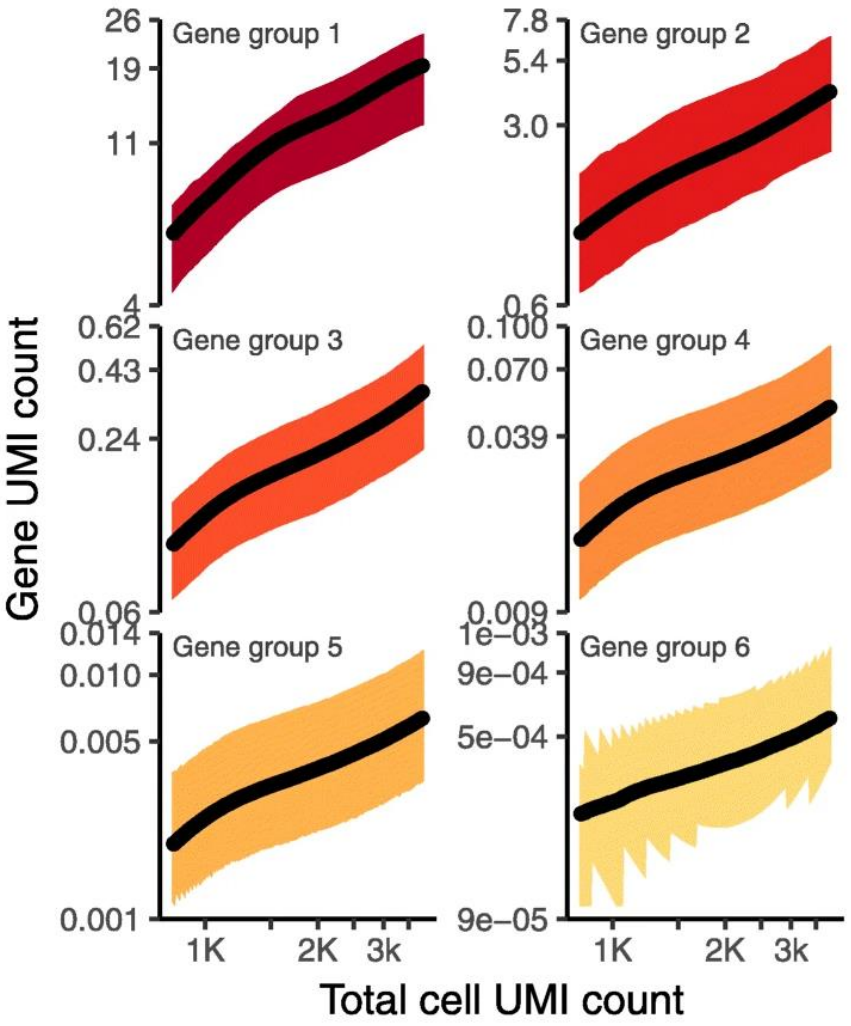
What other transformations can we use?

Does anyone know about `SCTransform`?

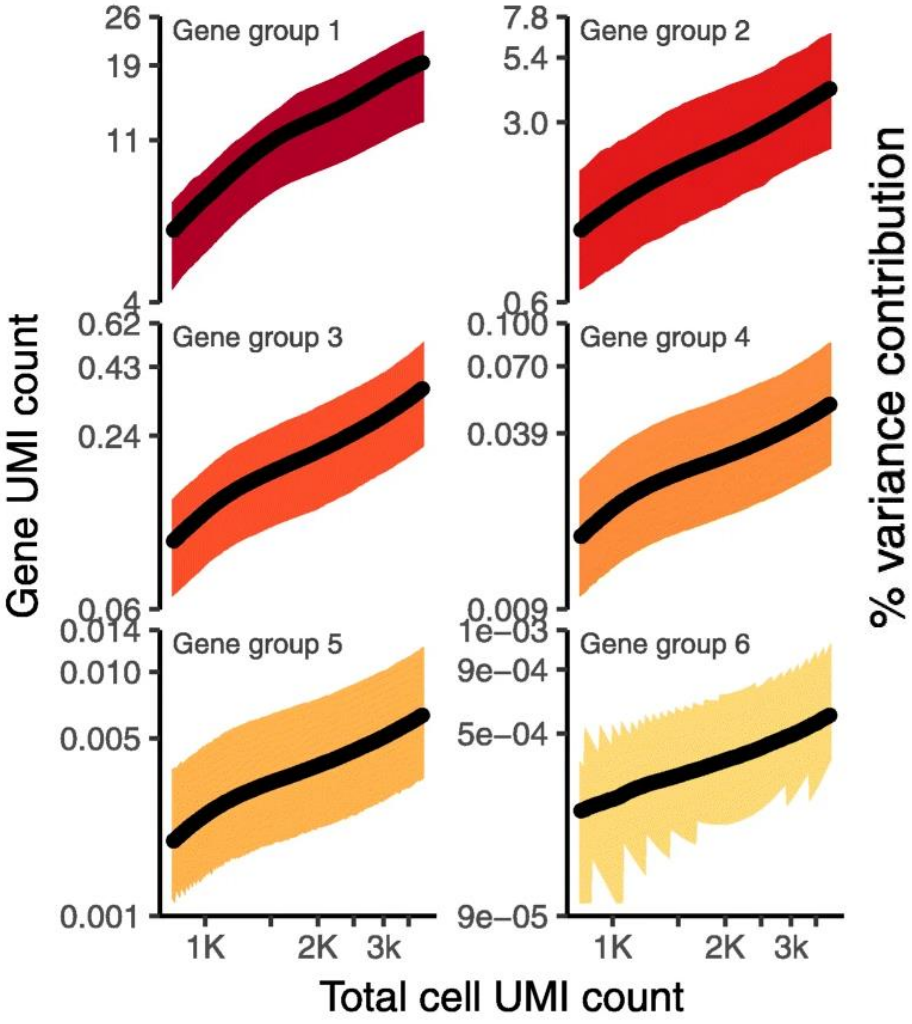
Transformation : Pearson residuals

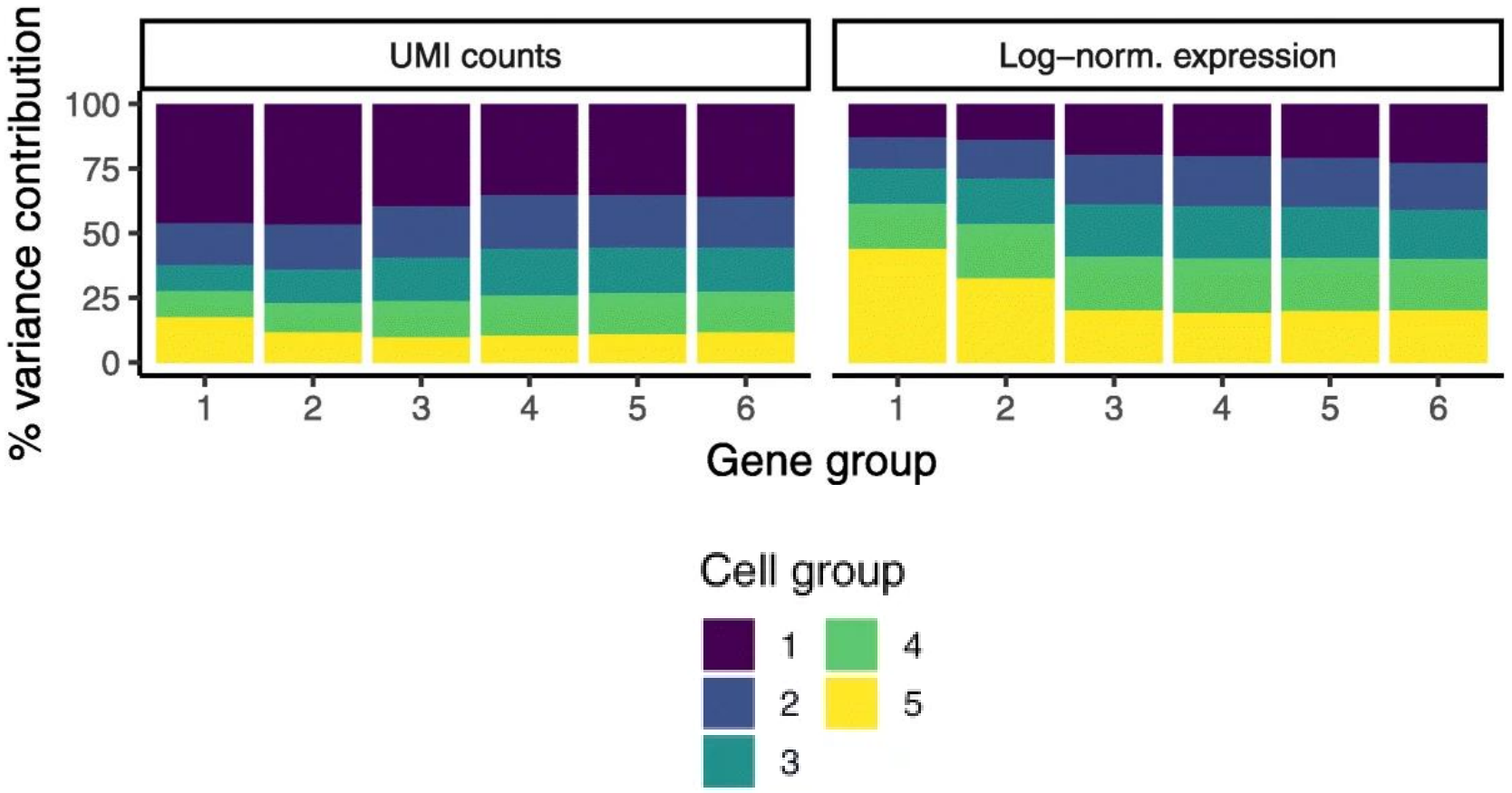
Instead of transforming each measurements individually, Pearson residuals apply a weight to all measurements of a gene

Gene counts are heavily influenced by sequencing depth



Gene counts are heavily influenced by sequencing depth





SCTransform: high or low for this cell?

Think of a child's growth chart

We don't just ask "how tall is this child?"
We ask "how tall for their age?"

The same height can mean very different things at different ages.

Growth chart	SCTransform
Child's age	How deeply the cell was sequenced
Child's height	How often a gene was detected
Growth curve	What we expect for that depth
Gap from the curve	**The score: how surprising it is**

Score = how far a count is from what we expect for that cell

Positive: higher than expected. Negative: lower. Close to zero: nothing unusual.

How SCTransform works, in three steps

1 Learn what's normal

For each gene, learn how often it is normally detected at each sequencing depth.

IN THE GROWTH CHART

Draw the curve of expected height for each age.

2 Share information

Rare genes are noisy, so borrow strength from genes with similar activity.

IN THE GROWTH CHART

Smooth the curve using many children, not just a few.

3 Score the surprise

Compare each count with what is expected for that cell. The gap becomes the score.

IN THE GROWTH CHART

Measure how far each child is from the curve.

Result: a depth-free score for every gene in every cell

Example: same count, different meaning

Gene X is normally detected about 1.5 times per 1,000 UMIs sequenced. **Both cells show 10 detections of gene X.**

	Cell A	Cell B
Sequencing depth	Shallow	Deep
Observed count	10	10
Expected count	3	15
SCTransform score	**+3.5**	**−0.8**

Cell A: surprisingly high

10 from a shallow cell is far above the 3 expected.

Cell B: slightly low

10 from a deep cell is a bit below the 15 expected.

Raw counts say the two cells are equal. SCTransform says they are not.

Illustrative numbers. The score is technically called a Pearson residual.

Example: abundant versus rare genes

Two genes in the same cell, both above expectation. Which jump matters more?

Gene	Expected count	Observed count	Raw difference	Score
LYZ	500	560	+60	+0.8
CD79A	0.5	3	+2.5	+3.5

+60 counts → +0.8

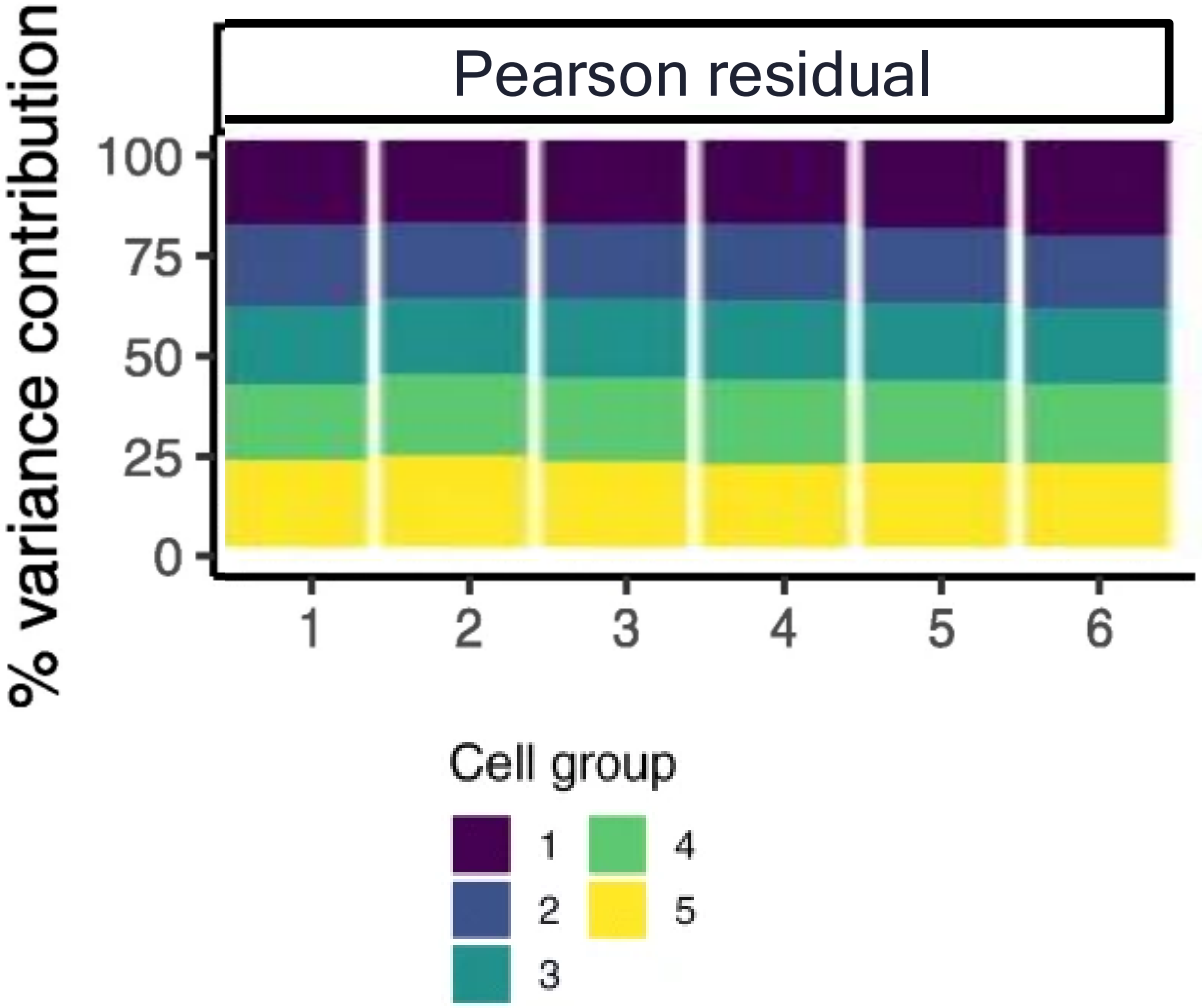
LYZ looks huge in raw data and would dominate PCA, but its score is small.

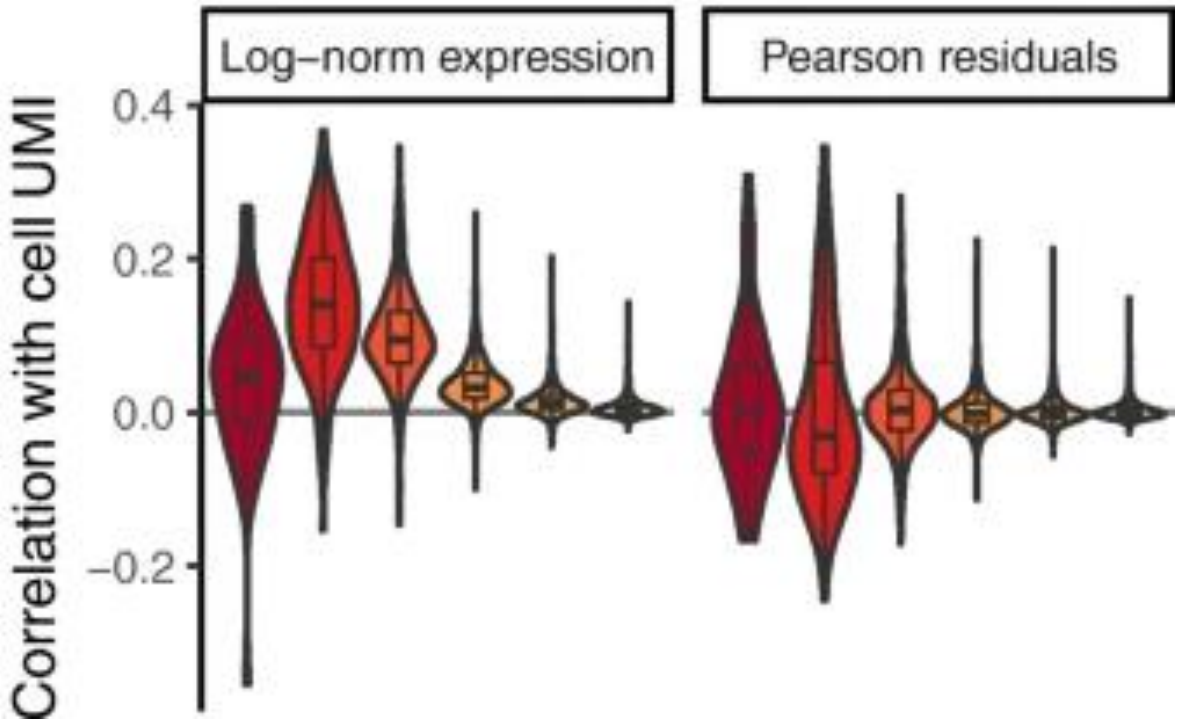
+2.5 counts → +3.5

CD79A looks tiny in raw data, but its score flags a real change.

Illustrative numbers. The score takes into account that abundant genes naturally fluctuate more than rare ones.

sctransform





We will not use `SCTransform` for the group work due to computation resources. It is advisable to evaluate it for your own single-cell RNA-seq data.

Are scaled values appropriate for downstream analyses like differential expression, and what are the reasons for or against their use?